

# The missing per-protein term is corpus context, not a property of the protein

Why a blind benchmark's pooled correlation falls so far below its per-protein correlation, and whether the gap can be closed · S669 and held-out Tsuboyama

**A per-protein additive offset is the missing correction, it is worth several times more across datasets than within one, and nothing we can extract predicts it.** An oracle offset lifts the leakage-clean S669 correlation from 0.453 to 0.643; an honest split-half estimate makes +0.144 of that transferable. The same correction is worth only +0.029 in-distribution. Homologous constructs share the per-protein mean  $\Delta\Delta G$  at  $r = 0.516$  but share the model's *error* on it at only  $r = 0.090$ . The model already extracts the part of the protein-level signal that the fold determines; what remains is assay and corpus context, which no representation of *the protein* can supply.

## 1. Motivation

On a diverse blind benchmark this predictor shows a large and consistent gap: it ranks mutations well *inside* a protein while placing different proteins poorly on a common scale. Pooled Pearson  $r$  sits far below the median per-protein  $r$  computed from the very same predictions. That gap has an obvious candidate explanation — a single wrong number per protein — and an obvious potential remedy: predict that number from the wild-type alone and add it back. If that worked it would be the cheapest large improvement available, since it needs no new training data, no architecture change and no per-variant computation.

Three questions follow, and they must be asked in order. What *kind* of correction is missing — an offset, a gain, or both? Is the apparent benefit real, or an artifact of fitting the correction on the same variants used to score it? And can the correction be predicted from anything available at inference time?

## 2. Methods

**Oracle ceiling.** Take existing predictions and apply the best possible per-protein offset, gain, and affine correction, then re-score. This bounds any scheme that predicts a per-protein number, however it is obtained.

**Honest split-half.** The oracle is fit and scored on the same variants, so it overstates. Estimating the offset from half of each protein's variants and scoring on the other half separates transferable signal from noise-fitting. Restricted to proteins with at least six variants, repeated over random splits.

**Predictability.** Four families of predictor were tried against the required offset: the wild-type Boltz embedding pooled over mutated positions; a head trained on all 550 proteins of both training corpora; interpretable descriptors (length, amino-acid composition, burial, hydropathy); and — the physically motivated candidate — the absolute folding free energy  $\Delta G$  of the wild type. Every predictor is evaluated under leave-one-protein-out or homology-grouped cross-validation, and judged against a constant baseline, because a target with small spread is trivially "predicted" by its own mean.

**The control that decides the question.** Constructs of the same base structure that differ by one background substitution let us ask, separately, whether the *quantity* (the protein's mean  $\Delta\Delta G$ ) and the model's *error* on that quantity are shared between close relatives. A fold property should be shared; corpus context should not.

**In-distribution comparison.** The whole analysis is repeated on held-out predictions within a single corpus, where train and test share assay conventions and curation. The contrast between the two regimes is the experiment's result.

## 3. Results

### 3.1 The missing correction is an offset, not a gain

| regime         | baseline | oracle offset | oracle gain | oracle affine | RMSE, gain  |
|----------------|----------|---------------|-------------|---------------|-------------|
| A — Tsuboyama  | 0.361    | <b>0.612</b>  | 0.407       | 0.607         | 1.70 → 1.80 |
| B — FireProt   | 0.460    | <b>0.649</b>  | 0.469       | 0.632         | 1.58 → 1.69 |
| D — fine-tuned | 0.453    | <b>0.643</b>  | 0.472       | 0.631         | 1.62 → 1.72 |

Pooled Pearson  $r$  on the leakage-clean S669 subset, with each oracle correction applied. The last column shows what the gain correction does to RMSE, which is the reason it is rejected even where it nudges  $r$  upward.

The offset is worth between 0.190 and 0.251  $r$  in every regime. The gain correction is not a competitor: it moves  $r$  only slightly and makes RMSE worse in all three regimes, and the affine correction — offset and gain together — never beats the offset alone. Whatever the model is getting wrong at the protein level is a constant shift, not a mis-scaled slope.

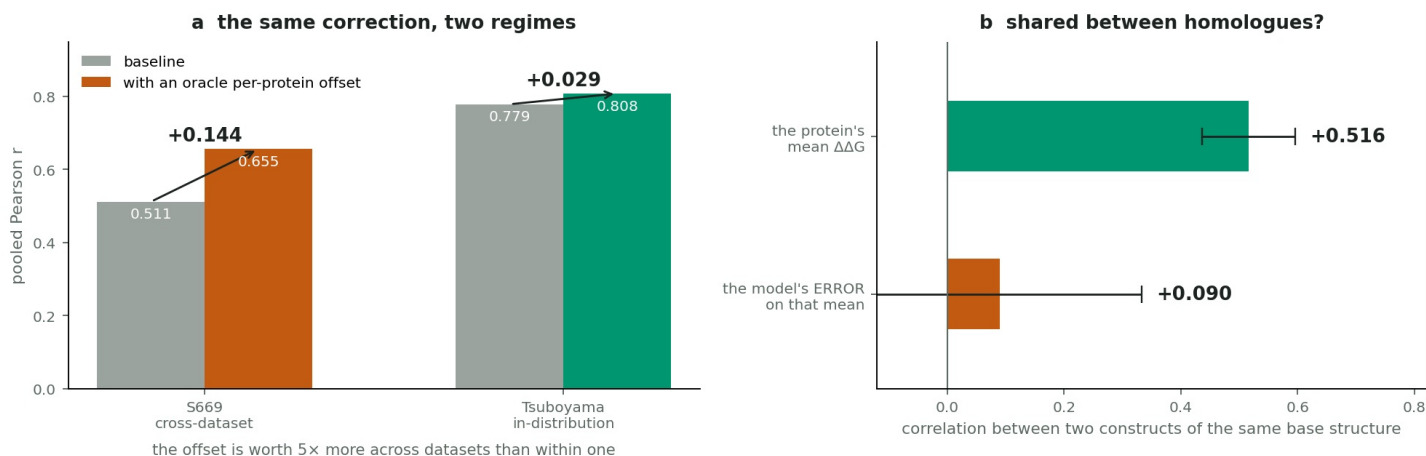
### 3.2 The benefit is real, and it is much larger across corpora than within one

Fitting and scoring the offset on the same variants inflates it. The honest split-half estimate on S669 moves the correlation from 0.511 to 0.655, a transferable gain of **+0.144**, with a further +0.055 attributable to noise-fitting. The quantity being estimated is stable rather than curation noise: the split-half reliability of the per-protein mean  $\Delta\Delta G$  is 0.823.

Repeating the identical procedure in-distribution gives a very different answer: 0.779 → 0.808, a gain of only **+0.029**. **The same correction is worth 5× more across datasets than within one**, and the offset it corrects is correspondingly smaller — a standard deviation of 0.29 kcal/mol in-distribution against 1.46 across corpora. The model is already well calibrated when train and test share a provenance.

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(a) A perfect per-protein offset buys far more when train and test come from different corpora than when they do not — so the model is already calibrated in-distribution. (b) Homologues share the quantity itself but not the model's error on it, so no representation of the protein can supply it.



**Figure 1.** (a) The same oracle correction applied in both regimes. (b) The control that decides the question: constructs of one base structure share the protein's mean  $\Delta\Delta G$ , but not the model's error on it.

### 3.3 Nothing available at inference time predicts the offset

**Not the wild-type embedding.** The best head reaches a correlation of +0.262 with the required offset, but applying its predictions never helps and usually hurts — on S669 the fine-tuned regime falls from 0.506 to 0.490 with a ridge head, and to 0.343 with a leave-one-protein-out head. In-distribution the same approach moves pooled  $r$  from 0.776 to 0.775, i.e. not at all.

**Not interpretable structure or chemistry.** Length, amino-acid composition, burial and hydropathy are all at or worse than a constant baseline. Length is the only descriptor with any univariate signal ( $|r| = 0.338$ ), and using it still degrades the pooled correlation.

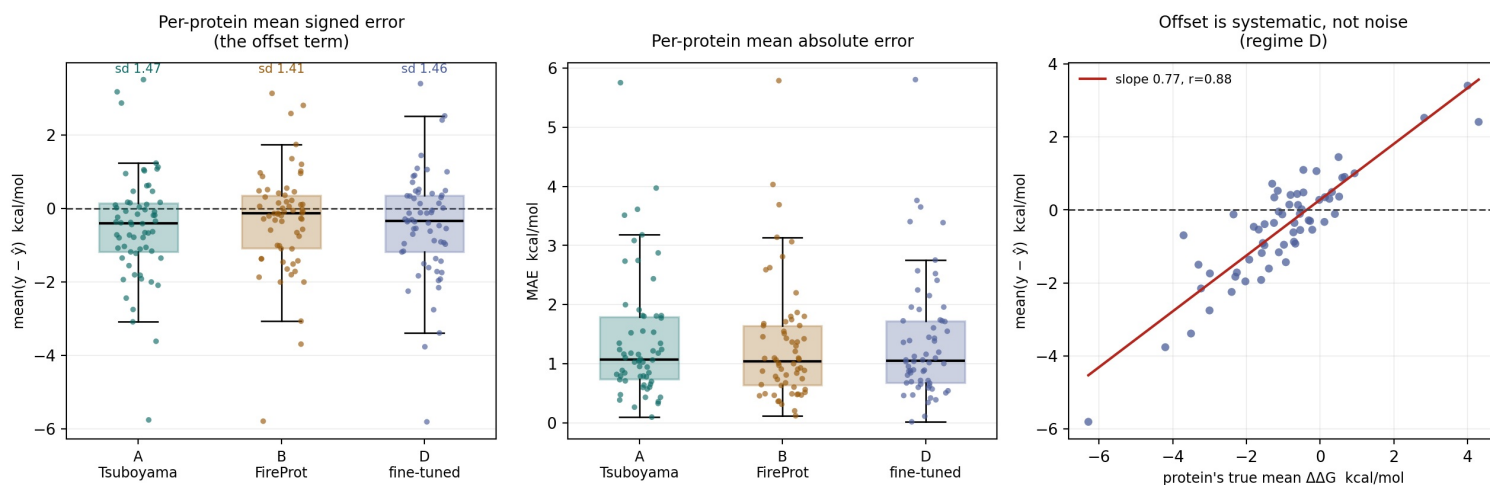
**Not the wild-type folding free energy.**  $\Delta G(\text{WT})$  is the physically motivated candidate for a per-protein term, and it is not recoverable from a frozen trunk either: whole-protein pooled representations predict it at  $r = 0.293$  under homology-grouped cross-validation, against  $r = 0.266$  from protein length alone.  $\Delta G(\text{WT})$  is itself only weakly a fold property — two constructs of the same base structure share it at  $r = 0.315$ , with a mean absolute difference of 0.64 kcal/mol against 1.07 for random pairs, roughly one mutation's worth of stability. Pooling a representation over residues averages away exactly what determines it.

### 3.4 The decisive control

Constructs of the same base structure share the **per-protein mean  $\Delta\Delta G$**  at  $r = 0.516 \pm 0.080$  — it is a genuine fold property. They share the model's **error** on that mean at only  $r = 0.090 \pm 0.242$ , indistinguishable from nothing.

That asymmetry is the answer. If the missing term were a property of the protein, the model's error on it would be shared between near-identical proteins, and a representation of the protein could in principle learn it. It is not shared. The model has already captured the part of the protein-level signal that the fold determines; the residual is corpus and assay context — which experimental laboratory, which curation lineage, which measurement convention — and no function of the protein alone can supply it.

S669 per-protein error — where the pooled correlation is lost (62 proteins, 541 variants)



**Figure 2.** Per-protein signed error and mean absolute error by training regime on the blind benchmark, and the per-protein error plotted against the protein's true mean  $\Delta\Delta G$ . The near-unit slope in the third panel says the model predicts close to the same mean for every protein regardless of what that protein's mean actually is.

## 4. Interpretation

This closes the per-protein-correction direction, and it reframes what the blind-benchmark deficit is. The gap between pooled and per-protein correlation is not a term the model failed to learn from its training data; it is cross-dataset calibration under domain shift. The distinction matters because the two diagnoses recommend opposite work. A missing learnable term would justify building a per-protein head. Domain shift does not — it points instead at calibration against a handful of labelled variants from the target assay, or at reporting the quantity the model actually estimates well.

That quantity is within-protein ranking, and it is where the defensible claim for this predictor lies. The per-protein correlation is high and stable across regimes and across homology filters; the cross-protein scale is where essentially all of the error lives.

A secondary result stands on its own: absolute folding stability is not recoverable from a frozen structure-prediction trunk by pooling its representation over residues. It is also only weakly determined by the fold, moving by about one mutation's worth of energy between constructs that differ by a single background substitution. Any scheme that hopes to read absolute  $\Delta G$  out of such a representation should expect the same null.

## 5. Limitations

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- The split-half analysis needs proteins with enough variants to halve, which restricts it to 19 of the benchmark's 62 proteins. It is the honest estimate available, but it is computed on the better-sampled subset, not the whole benchmark.
- The homology control uses constructs of the same base structure, of which there are only 39–51 groups depending on the quantity. The intervals are correspondingly wide, and the claim rests on the *gap* between the two correlations rather than on either value's precision.
- "Not predictable" is a statement about the predictors tried — pooled trunk representations, a trained head, interpretable descriptors, and  $\Delta G(\text{WT})$ . It does not exclude that some other feature of the wild type carries the term.
- The offset is defined against a specific benchmark's labels. Part of what it absorbs may be that benchmark's own curation idiosyncrasies rather than a property of the target assay in general — which is consistent with the domain-shift reading but not separable from it here.

## 6. Conclusion

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Asked whether the large gap between within-protein and cross-protein accuracy can be closed by supplying one number per protein, the answer is that the number exists, that a perfect version of it would help substantially on a blind benchmark and barely at all in-distribution, and that it is not predictable from the protein. The homology control explains why: the quantity is a fold property, but the model's error on it is not. What is missing is not information about the protein — it is information about the experiment.